

Executive Financial Performance Analysis

1. Executive Summary

This project provides a comprehensive end-to-end financial analysis of sales, revenue, cost structure, and profitability. The process involved data cleaning and Exploratory Data Analysis (EDA) in Python, data querying logic in SQL, and the development of an interactive executive dashboard in Power BI Desktop. The goal is to deliver actionable insights to optimize profitability across global markets and segments.

2. Python EDA & Data Cleaning Code

Data health verification and initial exploration were completed using Python's pandas library.

```
import pandas as pd

# Load the downloaded Financial Sample dataset
df = pd.read_excel('Sample data (1).xlsx')

# Standardize column names (strip extra spaces)
df.columns = df.columns.str.strip()

# Check dataset overview and data types
print("Dataset Shape:", df.shape)
print("\nMissing Values:")
print(df.isnull().sum())

# Display basic statistical summary
df.describe()
```

Key Data Cleaning Finding: The dataset contained 53 blank entries in Discount Band, which were confirmed to be regular transactions without discounts rather than corrupt data.

3. Key Financial Metrics (KPI Summary)

Total Units Sold: 1.13 Million

Total Revenue / Sales: \$118.73 Million

Total Cost of Goods Sold (COGS): \$102.00 Million

Total Net Profit: \$16.89 Million

4. Power BI Dashboard Architecture & Insights

An Executive Financial Dashboard was built featuring a dark navy header, rounded container cards, dynamic filters, and clean visualizations:

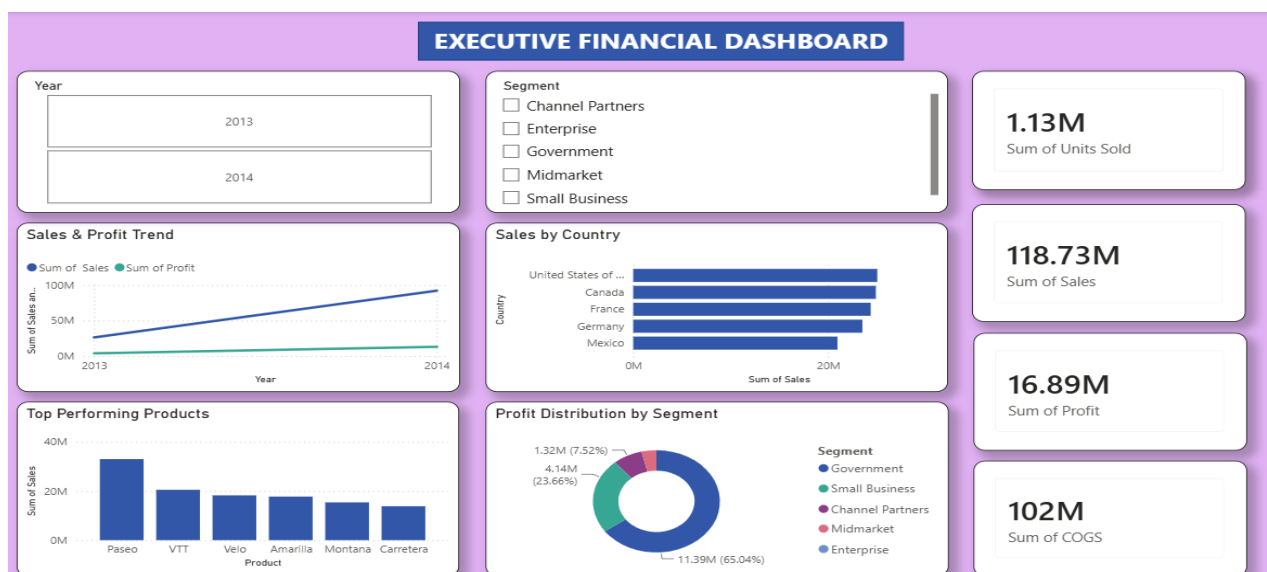
Sales & Profit Trend (Line Chart): Showcases strong Year-over-Year (YoY) revenue and profit growth from 2013 to 2014.

Sales by Country (Clustered Bar Chart): Identifies the United States as the primary sales driver, followed by Canada, France, Germany, and Mexico.

Profit Distribution by Segment (Donut Chart): Highlights that the Government segment accounts for over 65% of net profits.

Top Performing Products (Stacked Column Chart): Demonstrates that Paseo is the top revenue-generating product.

Interactive Controls: Integrated Year (2013/2014) and Segment tile slicers for real-time cross-filtering.



5. Strategic Business Recommendations

Protect High-Margin Segments: Prioritize government contracts and enterprise relationships as they provide the highest profit margin stability.

COGS Optimization: Cost of Goods Sold accounts for ~\$102M out of \$118.73M revenue. Reviewing production costs will directly increase net profitability.

Geographic Expansion: Leverage high-performing sales playbooks from the US market to boost market share in developing regions like Mexico.